MARIAM MERZA

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SKILLS

- **Technical Skills:** Python, SQL, Excel, PowerBI, JavaScript, pandas, scikit-learn, TensorFlow, NumPy, data visualization, machine learning, sentiment analysis, forecasting.
- **Research & Analytical Skills:** Statistical analysis, data preprocessing, classification and regression, model evaluation, feature engineering, technical research and reporting.
- **Business & Communication:** Research presentations, cross-functional collaboration, technical documentation, data storytelling.
- **Languages:** English (fluent), French (limited proficiency).

EDUCATION

***Computer Science** Bachelors of Science (Hons)* — April 2026

Trent University – Oshawa, ON

- **Relevant Coursework:** Applied AI & Machine Learning, Artificial Intelligence, Philosophy of AI, AI in Information Systems, Data Mining.

PUBLICATIONS

- M. Merza, U. Obinwanne and W. Feng, "Reducing Financial Debt and Illiteracy in Canadian Populations Using Machine Learning Prediction Models," *2025 IEEE/ACIS 29th International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing (SNPD)*, Busan, Korea, Republic of, 2025, pp. 998-1000, doi: 10.1109/SNPD65828.2025.11253447.

WORK EXPERIENCE

Research Assistant – Data Analysis
Trent University

September 2025 – January 2026 / Oshawa, ON

- Conducted data analysis on social media data (YouTube, Meta, Reddit, etc.) and secured access to platform APIs.
- Analyzed large-scale social media datasets using sentiment analysis and preprocessing techniques to identify behavioral and engagement patterns.
- Assisted with categorization development, data classification and logistic regressions.
- Applied machine learning classification techniques to support data categorization and predictive analysis initiatives.

Undergraduate Researcher
Trent University

May 2025 – July 2025 / Oshawa, ON

- Developed and compared ARIMA, LSTM and hybrid forecasting models on Canadian financial datasets.
- Performed data preprocessing, feature engineering and model evaluation (precision, recall, F1, and MSE metrics).
- Co-authored a research paper detailing methodology, results, and recommendations for improving financial decision-making.
- Presented the finished research paper at the SNPD2025-Summer IV International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing.

PROJECTS

Financial Algorithm | *Python, pandas, scikit-learn, TensorFlow/Keras, time-series analysis, academic writing.*
May 2025 – July 2025

- Developed an end-to-end financial ML pipeline to detect anomalies, forecast expenses, and generate personalized financial advice using synthetic user data.
- Built and compared ARIMA, LSTM and hybrid models, with the LSTM achieving an MSE of 168K (94% lower than ARIMA) for time-series forecasting.
- Implemented Isolation Forest anomaly detection techniques to identify irregular spending behavior patterns.
- Visualized forecast outputs and financial trends using Matplotlib.

Cheese Fat Level Prediction | *Python, pandas, scikit-learn, Docker, Git*
April 2025

- Developed a machine learning pipeline to classify cheese fat levels using data preprocessing, feature engineering, and predictive modeling techniques.
- Automated data ingestion and preprocessing workflows in Python, reducing manual preparation time and improving reproducibility.
- Evaluated model performance using classification metrics and optimized preprocessing methods to improve prediction reliability.
- Containerized the application with Docker, enabling non-technical users to run the prediction pipeline efficiently across environments.
- Documented methodology and workflow to support usability, maintainability, and collaboration.

Social Media Sentiment Analysis Research | *Python, NLP, sentiment analysis, APIs, pandas*
September 2025 – January 2026

- Conducted large-scale social media data analysis using datasets collected from platforms including YouTube, Reddit, and Meta.
- Performed data preprocessing, cleaning, and sentiment analysis to identify behavioral and engagement trends across online discussions.
- Applied classification and logistic regression techniques to support data categorization and analytical research objectives.
- Secured and managed API access for social media data collection and integration workflows.
- Collaborated on research initiatives involving machine learning analysis, data interpretation, and reporting of findings.